substantial number of physicists the world over who are vigorously pursuing string theory know that they are taking a risk: that a lifetime of effort might result in an inconclusive outcome. Undoubtedly, significant theoretical progress will continue, but will it be sufficient to overcome present hurdles and yield definitive, experimentally testable predictions? Will the indirect tests we have discussed above result in a true smoking gun for string theory? These questions are of central concern to all string theorists, but they are also questions about which nothing can really be said. Only the passage of time will reveal the answers. The beautiful simplicity of string theory, the way in which it tames the conflict between gravity and quantum mechanics, its ability to unify all of nature's ingredients, and its potential of limitless predictive power all serve to provide rich inspiration that makes the risk worth taking.

These lofty considerations have been continually reinforced by the ability of string theory to uncover remarkably new physical characteristics of a string-based universe—characteristics that reveal a subtle and deep coherence in the workings of nature. In the language introduced above, many of these are generic features that, regardless of currently unknown details, will be basic properties of a universe built of strings. Of these, the most astonishing have had a profound effect on our ever evolving understanding of space and time.

Part IV. String Theory and the Fabric of Spacetime